

TEAM ACE PROJECT PRESENTATION

SMART DRAIN & HEATWAVE RESPONSE SYSTEM

An integrated urban climate management platform that uses real-time weather API forecasts and IoT sensor data to proactively address **heatwave droughts** (automated micro-irrigation) and **torrential rain** (smart pre-storm drainage) using a single rainwater storage tank.

PROJECT TITLE

Smart Drain & Heatwave System

TEAM

Team ACE



Extreme Summer Climate Hazards in Urban Environments

Modern cities face compound climate hazards from alternating heatwaves and flash floods.



1. Heatwave & Drought

- Extreme heat dries soil below 30% moisture.
- Heat island effect causes plant wilting & discomfort.
- Manual watering wastes municipal tap water.



2. Torrential Rain & Flooding

- Downpours exceed storm sewer capacity.
- Rainwater tanks fill to 100% and overflow into streets.
- Lack of pre-storm drainage leads to urban flooding.



3. Limits of Conventional Systems

- Existing drains operate as one-way reactive systems.
- Stormwater is dumped into sewers instead of stored.
- No integration between flood prevention & irrigation.



The Need for an Integrated Solution

Urban infrastructure requires a smart system capable of automatically switching between **storing rainwater for drought irrigation** and **pre-draining tanks before severe storms land**.

One Tank, Two Jobs: Bi-Directional Climate Management



JOB 1: Heatwave Micro-Irrigation (P1 Pump)

- **Trigger Condition:** Clear weather, Soil moisture < 30%, Temp $\geq 30^{\circ}\text{C}$, and Tank storage > 20%.
- **Control Action:** Activates **P1 Irrigation Pump** to deliver a 12-second pulse (~0.34 L) to the crop box.
- **Benefit:** Recycles stored rainwater to preserve urban crops without wasting tap water.



JOB 2: Pre-Storm Flood Drainage (P2 Pump)

- **Trigger Condition:** KMA Open API receives Heavy Rain Forecast AND Tank storage level > 20%.
- **Control Action:** Activates **P2 Pre-drain Pump** to empty the tank down to 10% before downpour hits.
- **Benefit:** Creates an empty storage buffer in advance, preventing urban sewer overload.

The Rainwater Circularity Loop

The torrential downpour that threatens street flooding is stored into the tank, becoming the exact water source that cools the city during the next heatwave.

1. ESP32 Edge Node & Local Hardware

- **Sensors:** Capacitive Soil Moisture (ADC1), DHT22 Temp/Humidity, HC-SR04 Ultrasonic Tank Level.
- **Actuators:** 2-Channel 5V Relay controlling P1 Irrigation Pump & P2 Pre-drainage Pump.
- **Execution Role:** Reads telemetry every 2 s, reports via MQTT JSON, executes commands with 15 s safety watchdog.

Telemetry Payload (ESP32 → Pi, 2 s)

```
{ "ts": 1786342072.4, "node": "ace-esp32", "soil_moisture_pct": 27.4, "ambient_temp_c": 34.2, "tank_level_pct": 43.1 }
```

2. KMA Weather Forecast Open API

- **Open API Polling:** Fetches short-term weather forecast every 10 min from KMA data portal.
- **Grid Conversion:** Converts lat/lon geographic coordinates to Lambert conformal conic grid.
- **Fail-Safe:** API outages default to "last valid forecast" then "no rain", preventing erroneous drainage.

Command Payload (Pi → ESP32, 5 s)

```
{ "mode": "GREENERY_IRRIGATION", "pump": true, "drain": false, "reason": "moisture 27%, 34.2C", "ttl_s": 15.0 }
```

System Decision Logic

CORE DECISION ENGINE

5-Second Real-Time Decision Algorithm

Evaluates sensor data and weather forecast every 5 seconds to automatically toggle between pre-storm flood protection and heatwave micro-irrigation.

Real-Time Sensor & Weather Polling

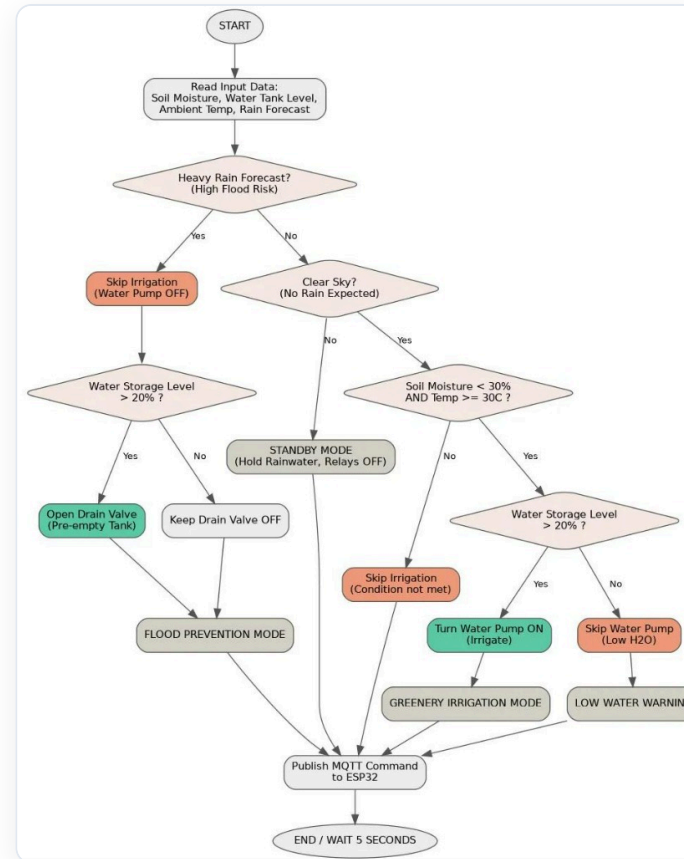
Polls KMA forecast API, soil moisture, ambient temp, and water tank storage level every 5-second cycle.

Heavy Rain Priority Override

Heavy rain forecast triggers instant pre-drainage via P2 pump, clearing tank storage capacity before storm hits.

Smart Micro-Irrigation

Doses a 0.34 L pulse via P1 pump during hot ($\geq 30^{\circ}\text{C}$) and dry ($< 30\%$ moisture) conditions when stored water is available.



Conventional Systems vs. Team ACE SDS Smart Platform

Feature	Conventional Municipal Drainage	ACE SDS Smart Platform
Primary Goal	Flood prevention only	Flood + Heatwave + Drought integrated control
Rainwater Processing	Store then discharge into sewer (wasted)	Filtered → Stored → 100% Recycled for Irrigation & Mist
Water Flow Mechanism	One-way gravity discharge	Bi-directional smart routing (P1 Irrigation & P2 Pre-drain)
Decision Timing	Reactive response after heavy rain	Proactive 1-hour KMA forecast API poller
Application Scope	Large public civil infrastructure	Roofs, planter boxes, smart shade micro-zones

 Smart Shade Cooling Fog Integration & Innovations

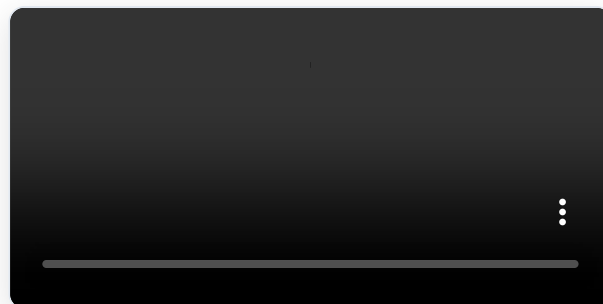
 **Rainwater First**
Prioritizes stored rainwater over tap water.

 **Multi-Height Spray**
Nozzles for children (1.2m) & adults (1.8m).

 **IoT Climate Control**
Lowers felt ambient temp by 3~5°C.

 **3-Tier Filtration**
Topsoil, fabric, & gravel prevent clogging.

1:1 Physical Prototype Live Demonstration Video



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Thank you for your time.

Question & Answer (Q&A)

We are ready for any questions or feedback.